

Task on AWS SQS, AWS MQ, AWS Secrets Manager, AWS WAF, & AWS Systems Manager.

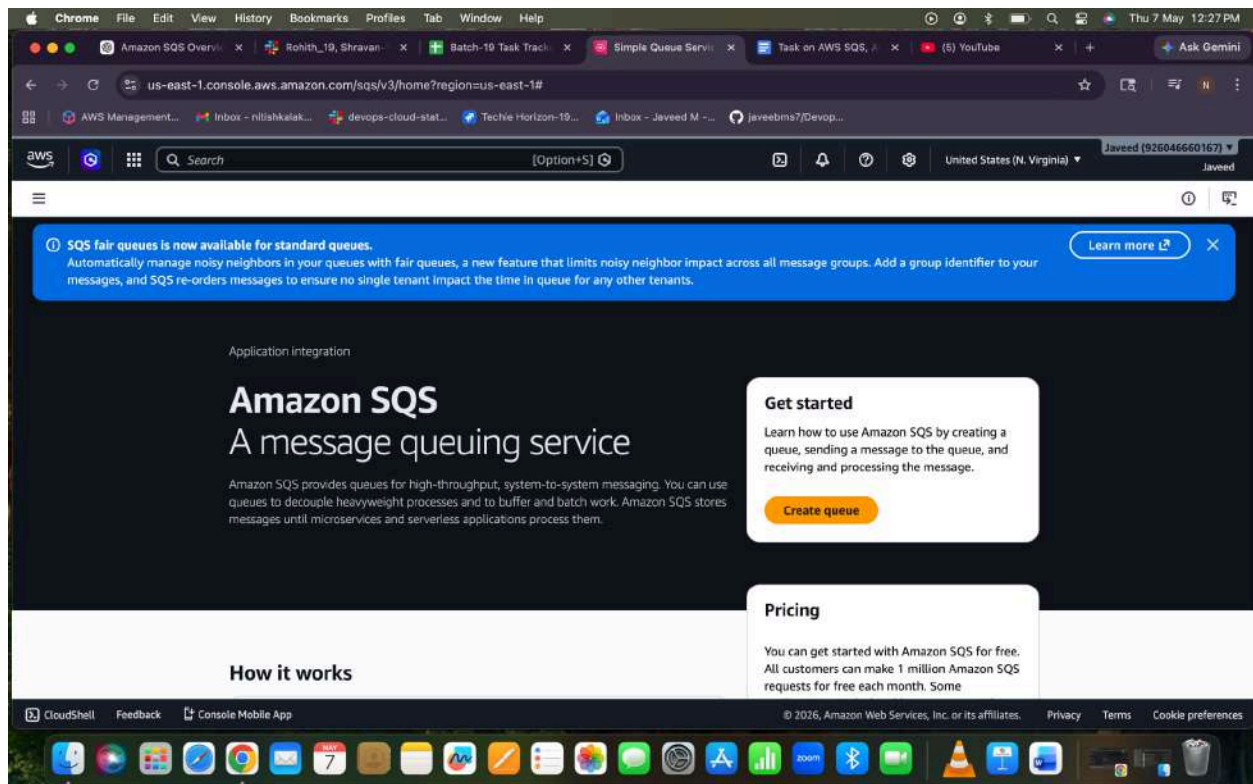
Amazon Simple Queue Service: SQS is a fully managed message queuing service in AWS used to send, store, and receive messages between applications or services.

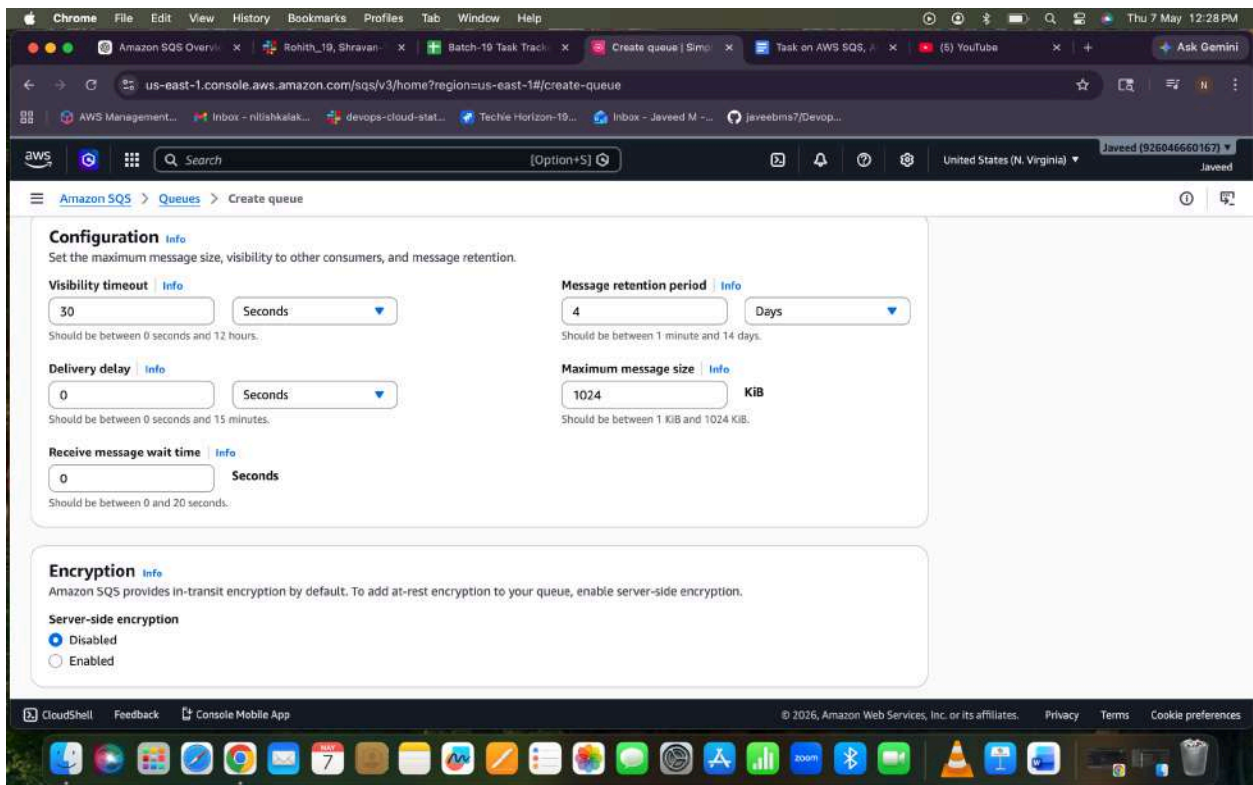
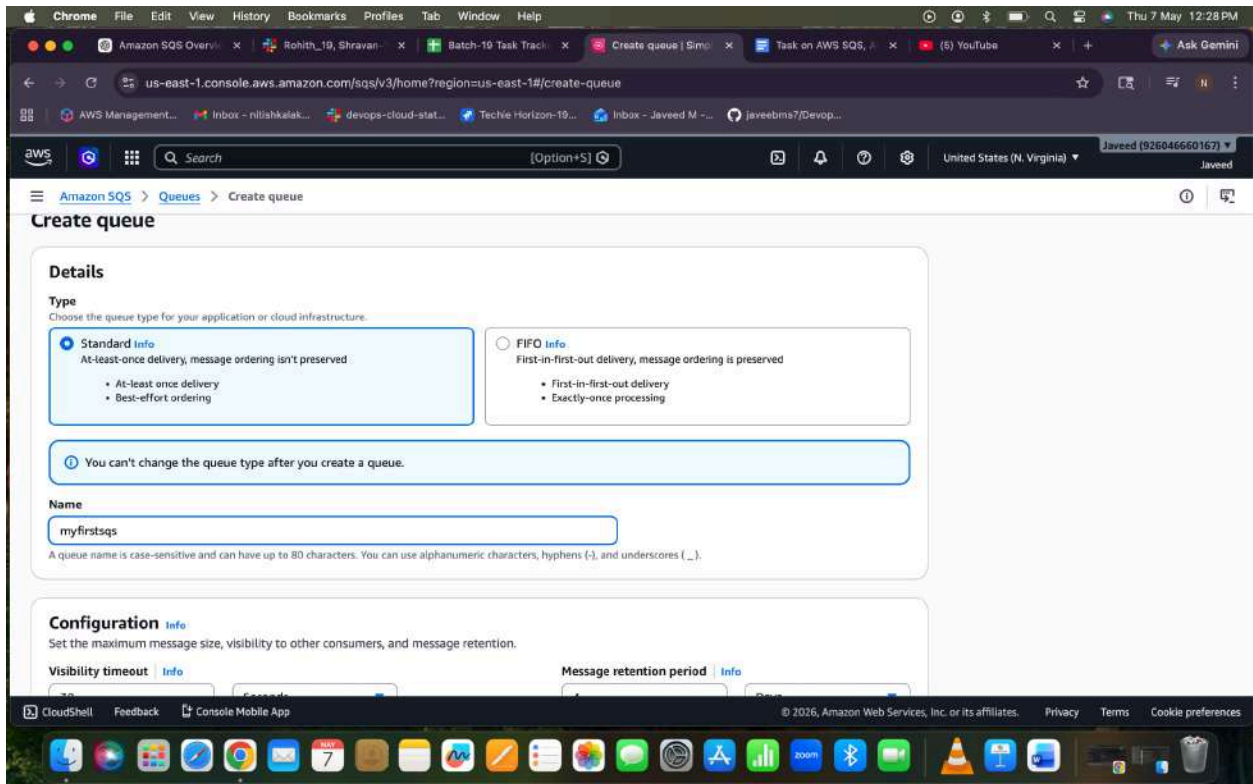
It helps different components of an application communicate asynchronously without directly depending on each other.

Think of it like a buffer/waiting line between two systems.

To create SQS

SQS → create → type(**Standard**/FIFO) → Name → configurations → Access Policy → Choose method(**Basic**/advanced) → Create





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Amazon SQS Overview Rohith_19, Shravan Batch-10 Task Track Create queue | Simp Task on AWS SQS, (5) YouTube Ask Gemini

us-east-1.console.aws.amazon.com/sqs/v3/home?region=us-east-1#/create-queue

AWS Management... Inbox - nitiashkalak... devops-cloud-stat... Techie Horizon-10... Inbox - Javeed M... javeebms7/Devop...

aws Search [Option+S] United States (N. Virginia) Javeed (926046660167) Javeed

Amazon SQS > Queues > Create queue

Choose method

☒ Basic
Use simple criteria to define a basic access policy.

☐ Advanced
Use a JSON object to define an advanced access policy.

Define who can send messages to the queue

☒ Only the queue owner
Only the owner of the queue can send messages to the queue.

☐ Only the specified AWS accounts, IAM users and roles
Only the specified AWS account IDs, IAM users and roles can send messages to the queue.

Define who can receive messages from the queue

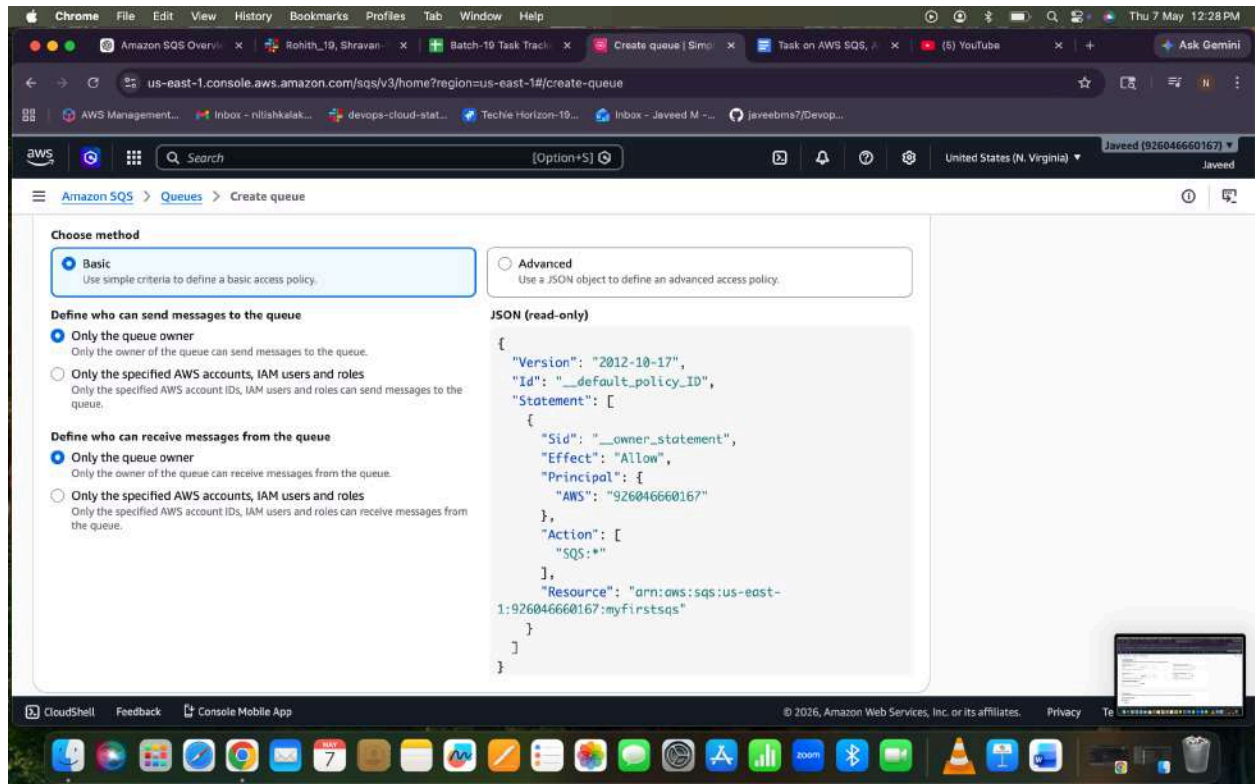
☒ Only the queue owner
Only the owner of the queue can receive messages from the queue.

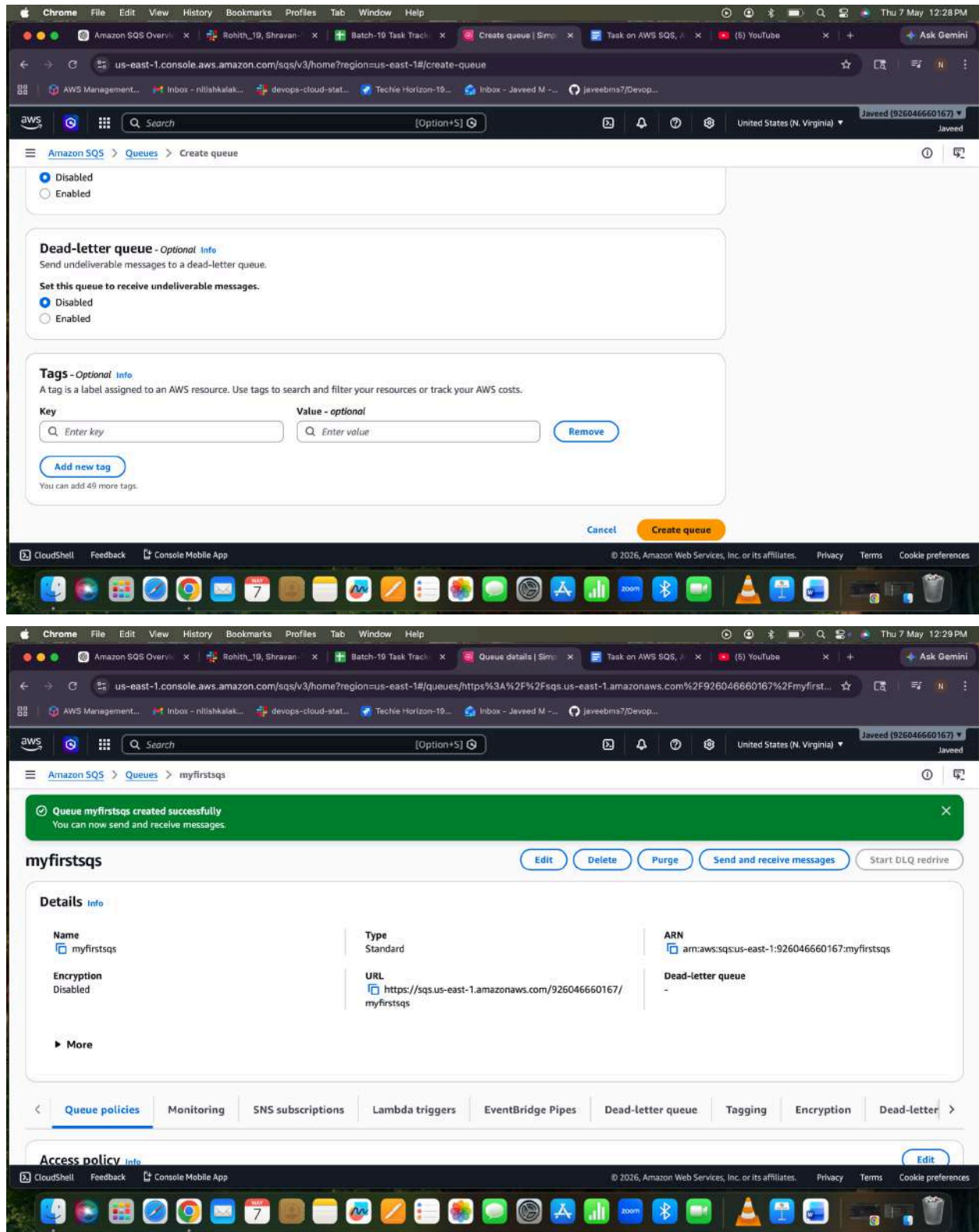
☐ Only the specified AWS accounts, IAM users and roles
Only the specified AWS account IDs, IAM users and roles can receive messages from the queue.

JSON (read-only)

```
{
  "Version": "2012-10-17",
  "Id": "__default_policy_ID",
  "Statement": [
    {
      "Sid": "__owner_statement",
      "Effect": "Allow",
      "Principal": {
        "AWS": "926046660167"
      },
      "Action": [
        "SQS:*"
      ],
      "Resource": "arn:aws:sqs:us-east-1:926046660167:myfirstsqs"
    }
  ]
}
```

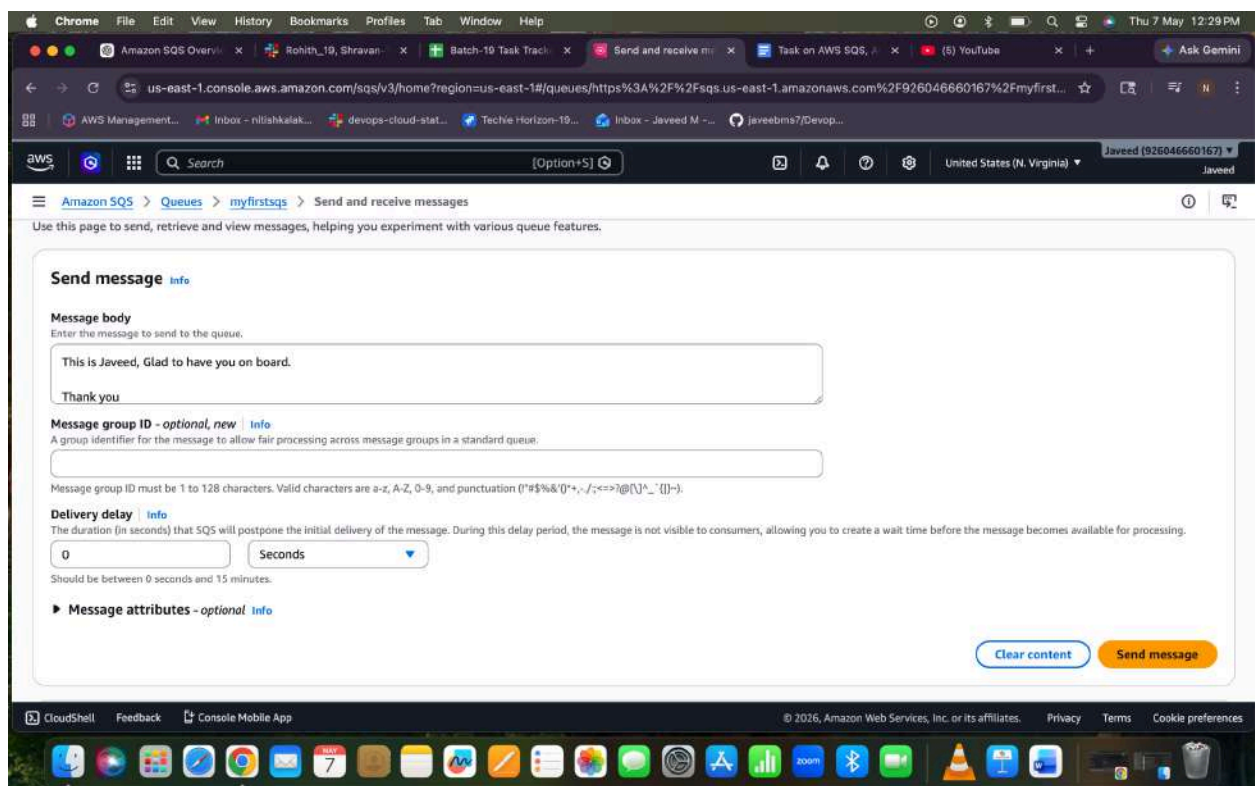
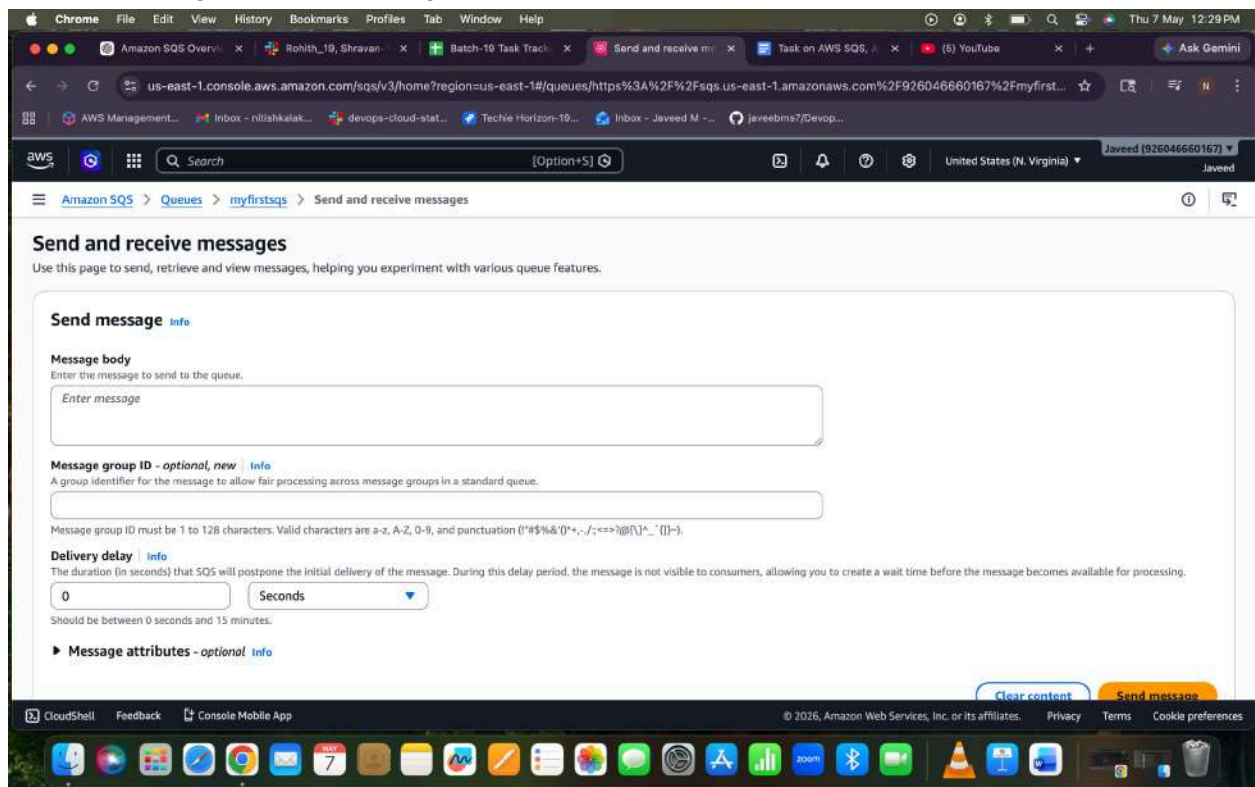
CloudShell Feedback Console Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Te

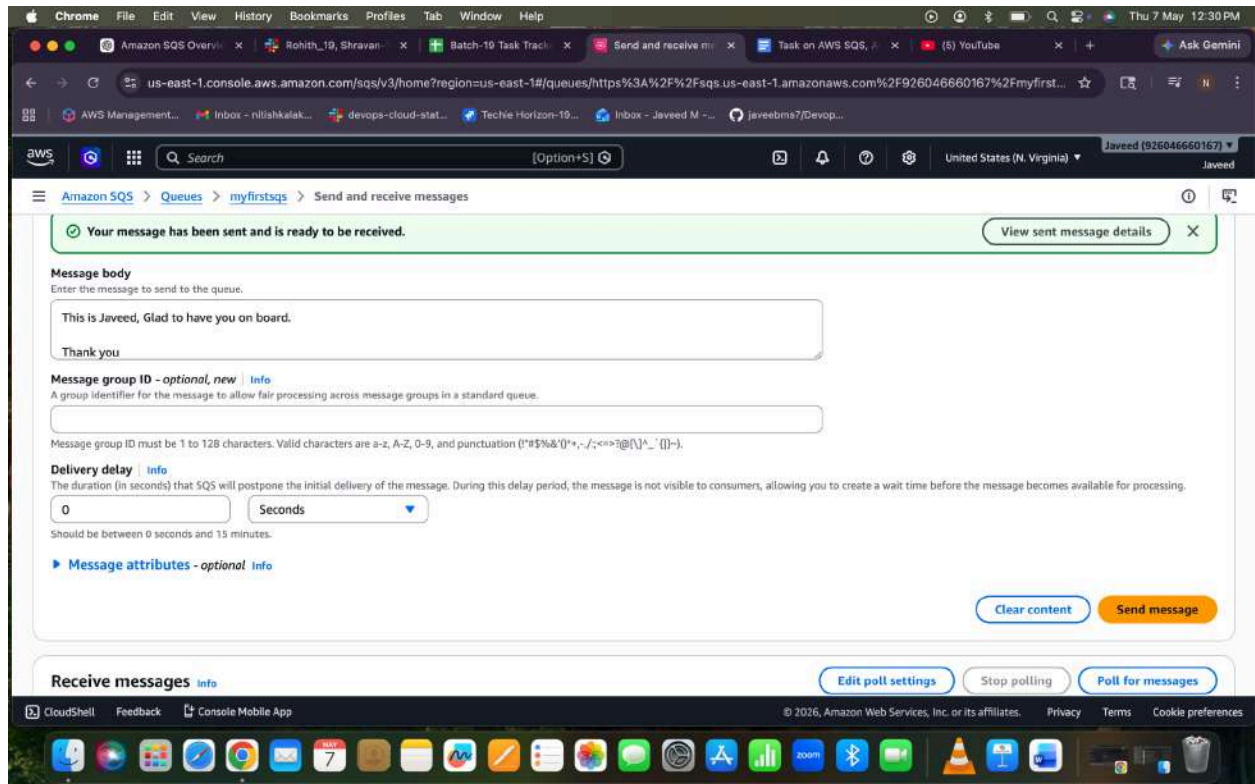




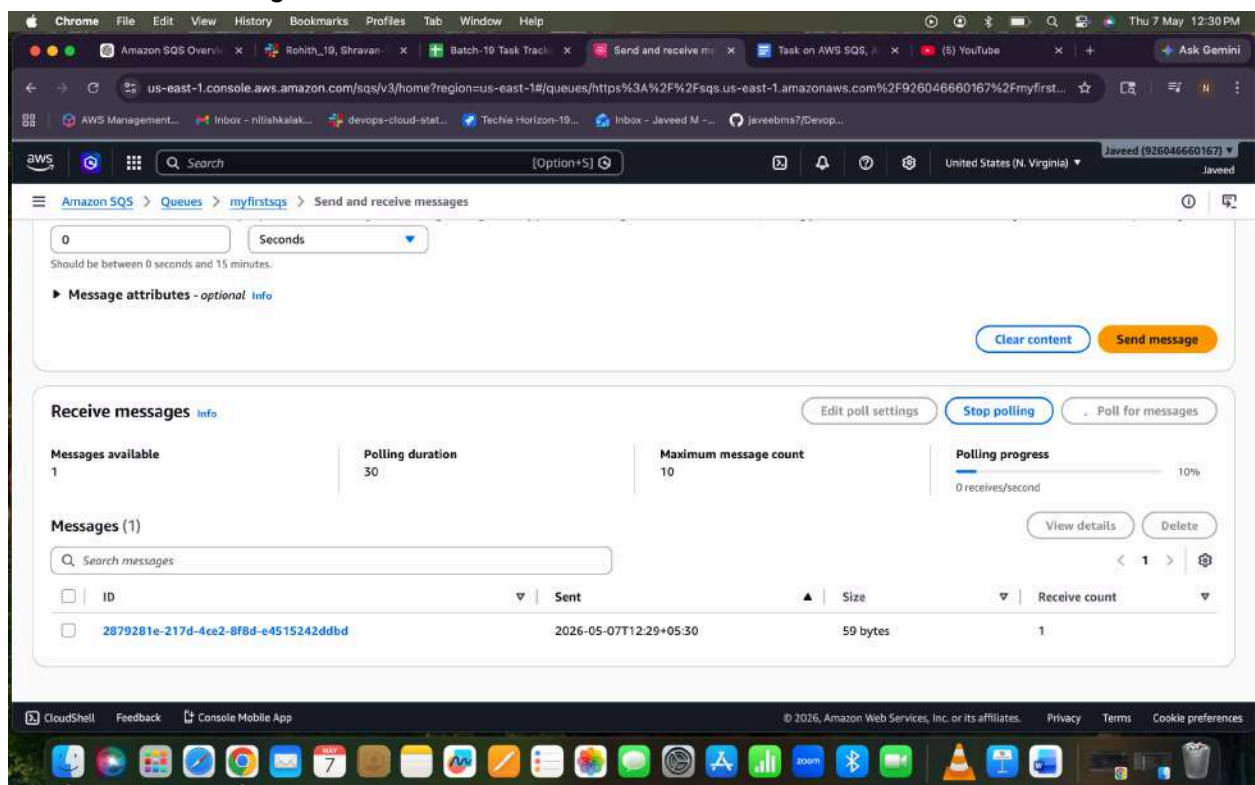
For Testing to send and receive messages

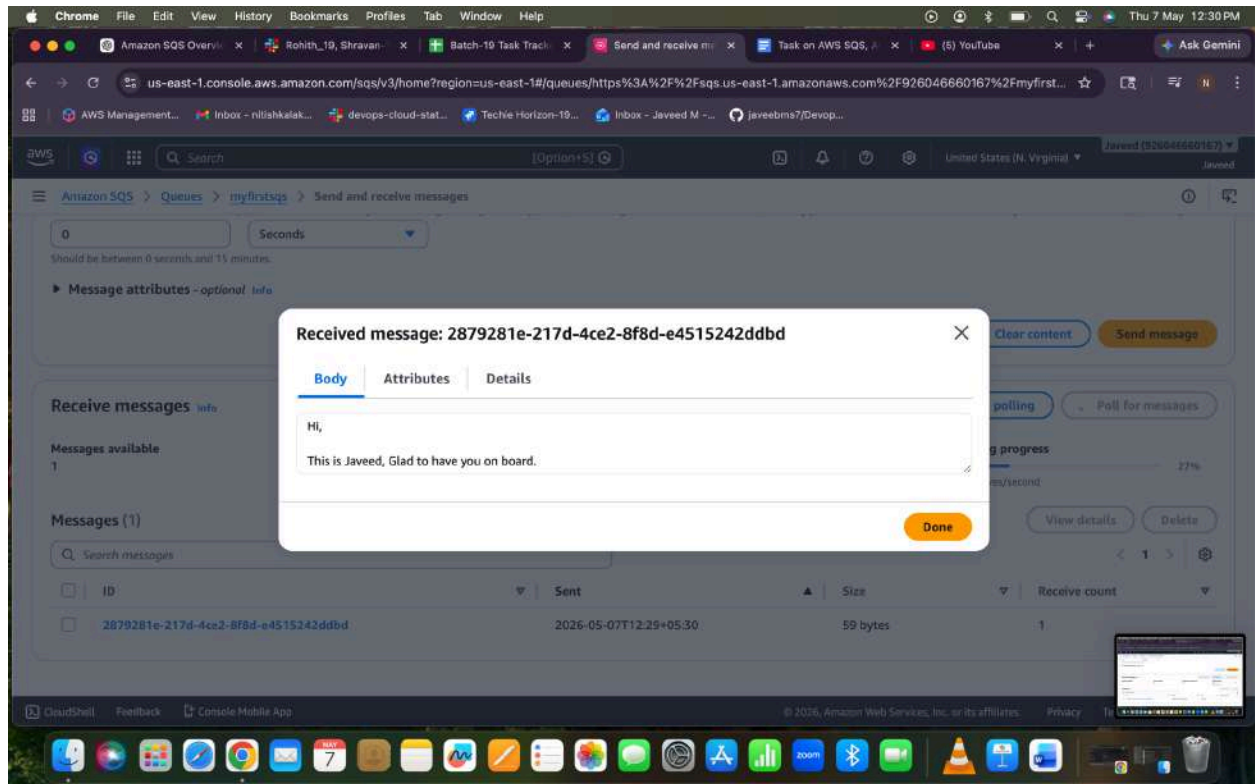
SQS→Queues→My Queues→Send & Receive Messages→Send message Info→Send Message→ Receive Message→Poll Message(to receive message) →Delete(To delete and receive message and stop polling).



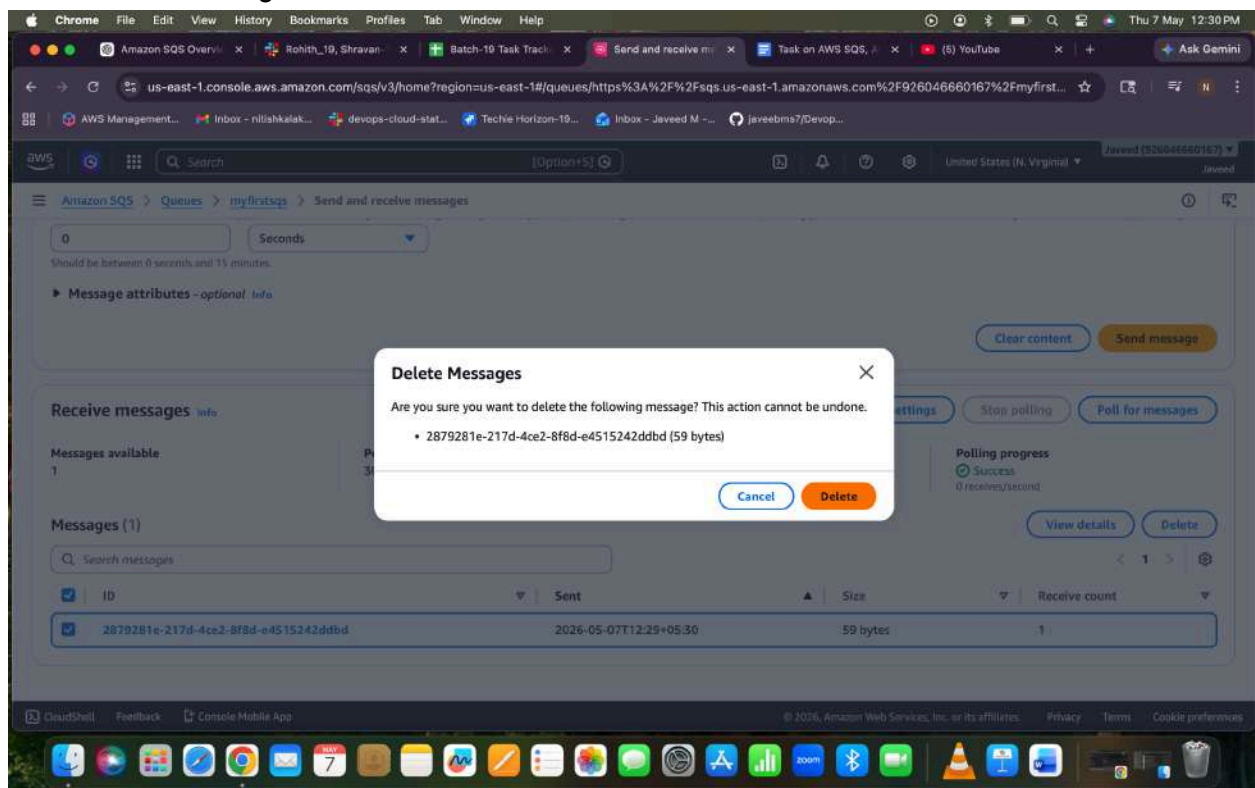


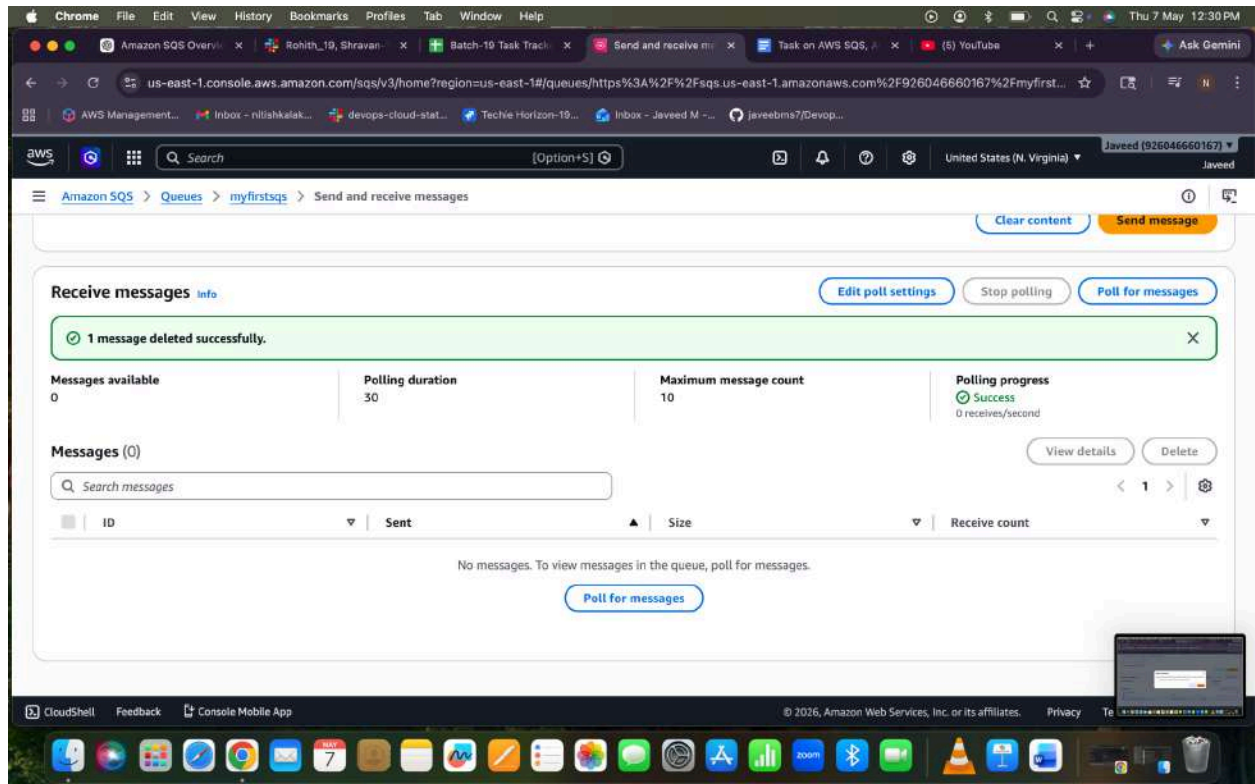
Received the message





Deleted The Message





AWS MQ:

Amazon MQ is a fully managed message broker service provided by AWS.

It is mainly used when applications need to communicate using traditional messaging protocols like:

- JMS
- AMQP
- MQTT
- STOMP
- OpenWire

AWS manages:

- Server setup
- Patching
- Backups
- High availability
- Monitoring

so we can focus on the application.

Amazon MQ→Brokers→Create Active MQ brokers→Select Broker Engine(Apache ActiveMQ/**Rabbit MQ**) →Deployment Mode(**Single InstanceBroker**/Cluster Deployment)
→Configure Settings→Details→Broker Name→ Instance Type→Rabbit MQ Access
→Username→password→create

us-east-1.console.aws.amazon.com/amazon-mq/home?region=us-east-1#/brokers/details?id=b-32a9d6af-8d33-43eb-88e5-d9c163d9dfcf

Amazon MQ

Brokers

Configurations

mydemo

Broker mydemo is being created. Broker creation takes about 20 minutes. Your broker's configuration can be found on the Configurations page.

Details

Specifications

ARN [info](#)

arn:aws:mq:us-east-1:926046660167:broker:mydemo-b-32a9d6af-8d33-43eb-88e5-d9c163d9dfcf

Broker name

mydemo

Broker status

Running

Broker instance type [info](#)

mq.m7g.medium

Configuration

Configuration name

mydemo-configuration

Configuration revision

Revision 1 - Auto-generated default for mydemo-configuration on RabbitMQ 4.2 [info](#)

CloudWatch Logs

General

Disabled - [Logs](#)

Security and network

VPC [info](#)

vpc-08bc924a2061938df [info](#)

Subnet(s) [info](#)

subnet-0d2e47f3b5fba65ea [info](#)

Public accessibility [info](#)

Yes

Maintenance

Automatic minor version upgrade

Yes

Maintenance window

Friday 22:00 - 00:00 UTC

us-east-1.console.aws.amazon.com/amazon-mq/home?region=us-east-1#/brokers/details?id=b-32a9d6af-8d33-43eb-88e5-d9c163d9dfcf

Amazon MQ

Brokers

Configurations

mydemo

Broker mydemo is being created. Broker creation takes about 20 minutes. Your broker's configuration can be found on the Configurations page.

Connections

Access your queues and exchanges and connect your application to the broker. If you disable public accessibility for your broker, your endpoints are reachable only within a VPC.

RabbitMQ web console

<https://b-32a9d6af-8d33-43eb-88e5-d9c163d9dfcf.mq.us-east-1.on.aws>

Endpoints

AMQP

<https://b-32a9d6af-8d33-43eb-88e5-d9c163d9dfcf.mq.us-east-1.on.aws:5671>

Prometheus

<https://b-32a9d6af-8d33-43eb-88e5-d9c163d9dfcf.mq.us-east-1.on.aws:9090>

Tags (0)

Manage tags

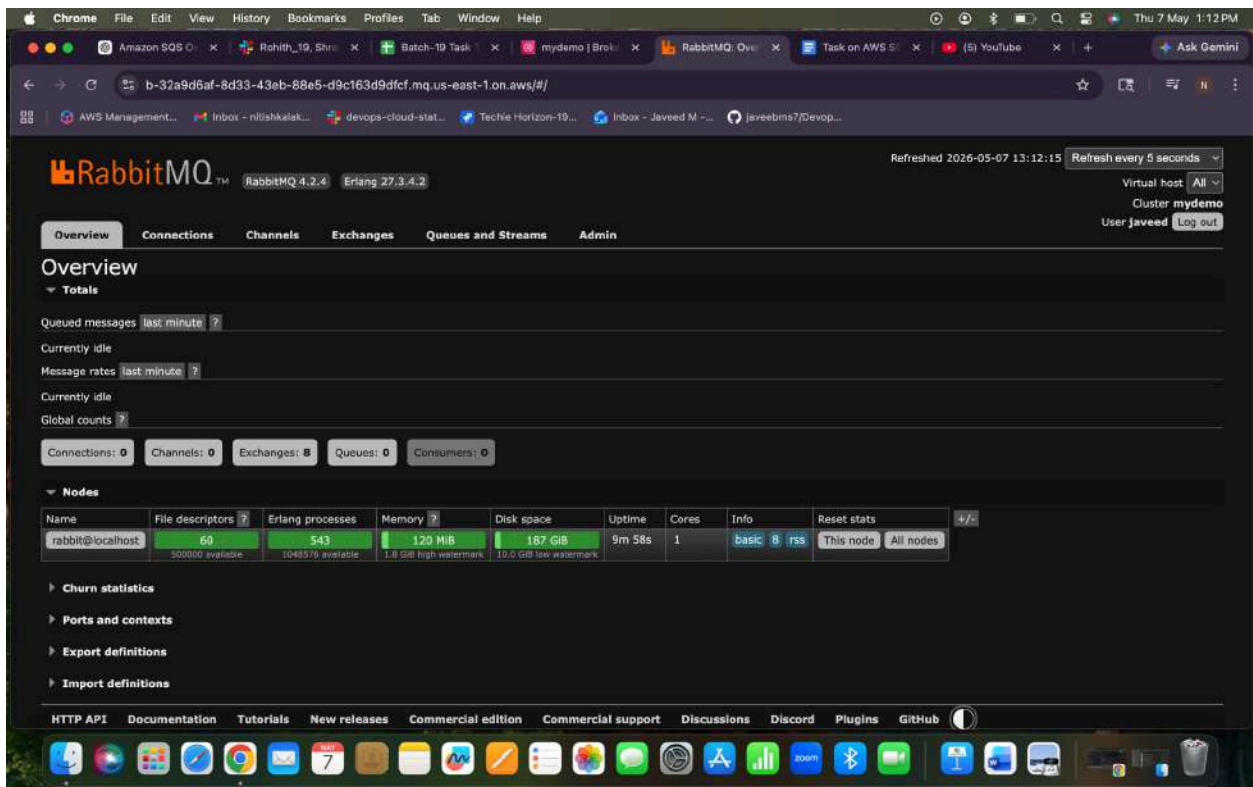
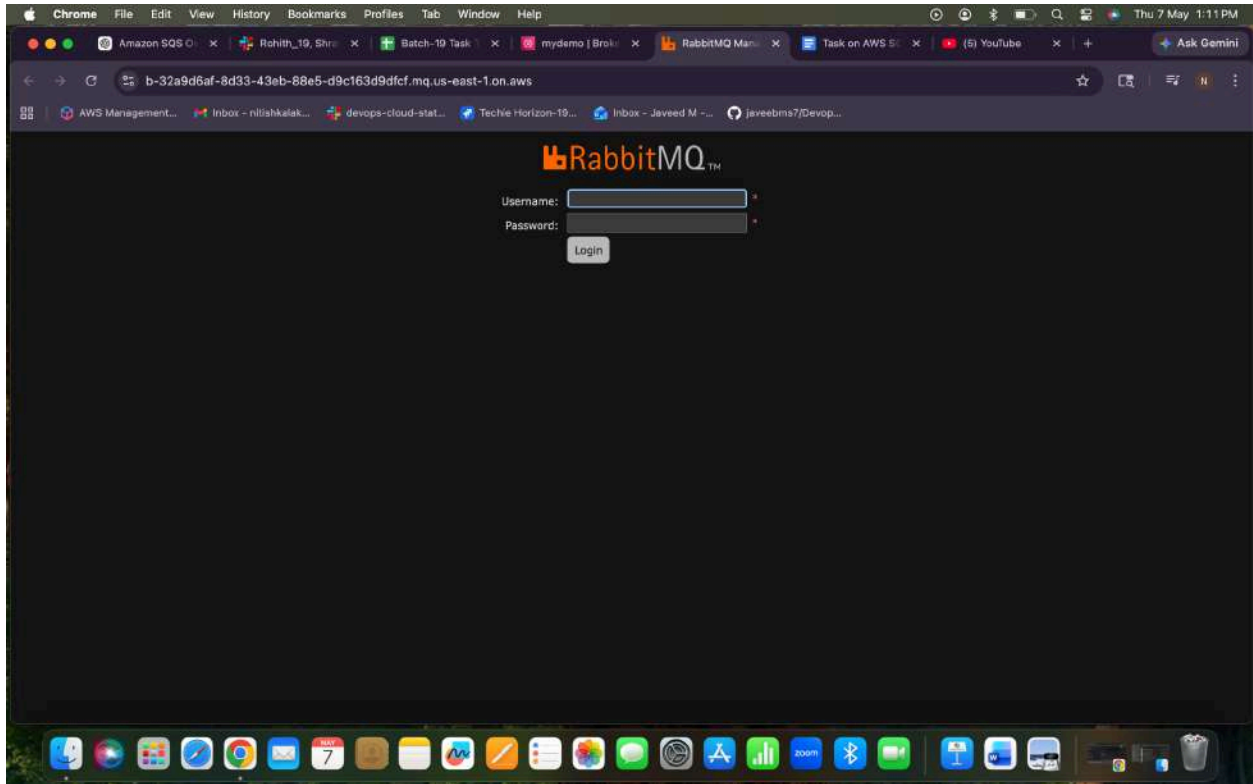
Key

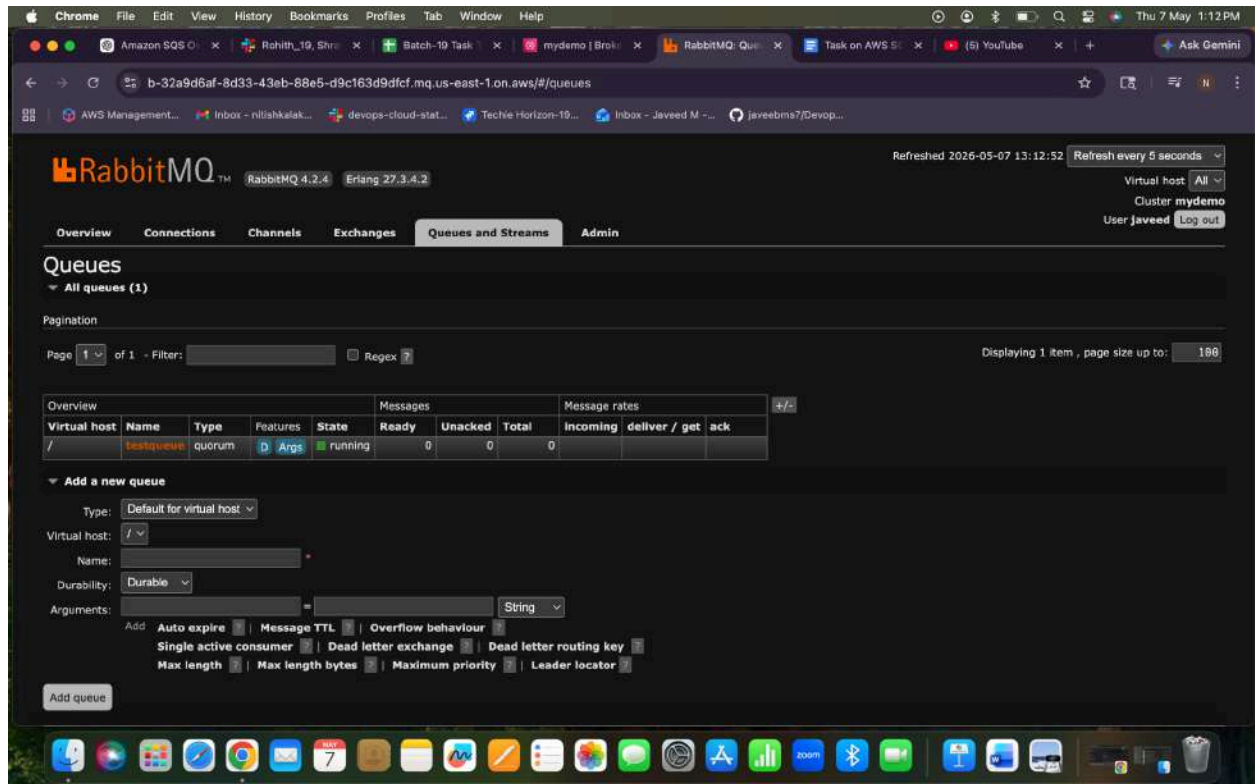
Value

No tags

To redirect to RabbitMQ console

Amazon MQ→Brokers→My brokers→Connections→RabbitMQ web console→URL Link→Login





To add a queue and publish a message and GET the message

Amazon MQ→Brokers→My brokers→Connections→RabbitMQ web console→URL

Link→Login→Queues & Streams →Queues→Add a new Queue→Name→Add Queue→ New Queue created→open→ Publish message→pay load(give any message →Publish message→ GET Message→ GET messages→Delete

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Amazon SQS x Rohith_19, Shr x Batch-19 Task x mydemo | Brok x RabbitMQ Que x Task on AWS S x (5) YouTube x + Ask Gemini

b-32a9d6af-8d33-43eb-88e5-d9c163d9dcf.mq.us-east-1.on.aws/#/queues/%2F/testqueue

AWS Management... Inbox - nitiashkalak... devops-cloud-stat... Techie Horizon-T9... Inbox - Javeed M... jareebms7/Devop...

RabbitMQ™ RabbitMQ 4.2.4 Erlang 27.3.4.2 Refreshed 2026-05-07 13:13:05 Refresh every 5 seconds Virtual host All Cluster mydemo User Javeed Log out

Overview Connections Channels Exchanges **Queues and Streams** Admin

	Total	Ready	Unacked	High priority	Normal priority	Returned	Dead-lettered
Messages	0	0	0	0	0	0	0
Message body bytes	0 B	0 B	0 B				0 B
Process memory	42 KIB						

Consumers (0)

Bindings (1)

Publish message

Message will be published to the default exchange with routing key **testqueue**, routing it to this queue.

Headers: = String

Properties: =

Payload:

Payload encoding: String (default)

Chrome File Edit View History Bookmarks Profiles Tab Window Help Thu 7 May 1:13 PM

Amazon SQS x Rohith_19, Shr x Batch-19 Task x mydemo | Brok x RabbitMQ Que x Task on AWS S x (5) YouTube x + Ask Gemini

b-32a9d6af-8d33-43eb-88e5-d9c163d9dcf.mq.us-east-1.on.aws/#/queues/%2F/testqueue

AWS Management... Inbox - nitiashkalak... devops-cloud-stat... Techie Horizon-T9... Inbox - Javeed M... jareebms7/Devop...

RabbitMQ™ RabbitMQ 4.2.4 Erlang 27.3.4.2 Refreshed 2026-05-07 13:13:25 Refresh every 5 seconds Virtual host All Cluster mydemo User Javeed Log out

Overview Connections Channels Exchanges **Queues and Streams** Admin

	Total	Ready	Unacked	High priority	Normal priority	Returned	Dead-lettered
Messages	0	0	0	0	0	0	0
Message body bytes	0 B	0 B	0 B				0 B
Process memory	34 KIB						

Consumers (0)

Bindings (1)

Publish message

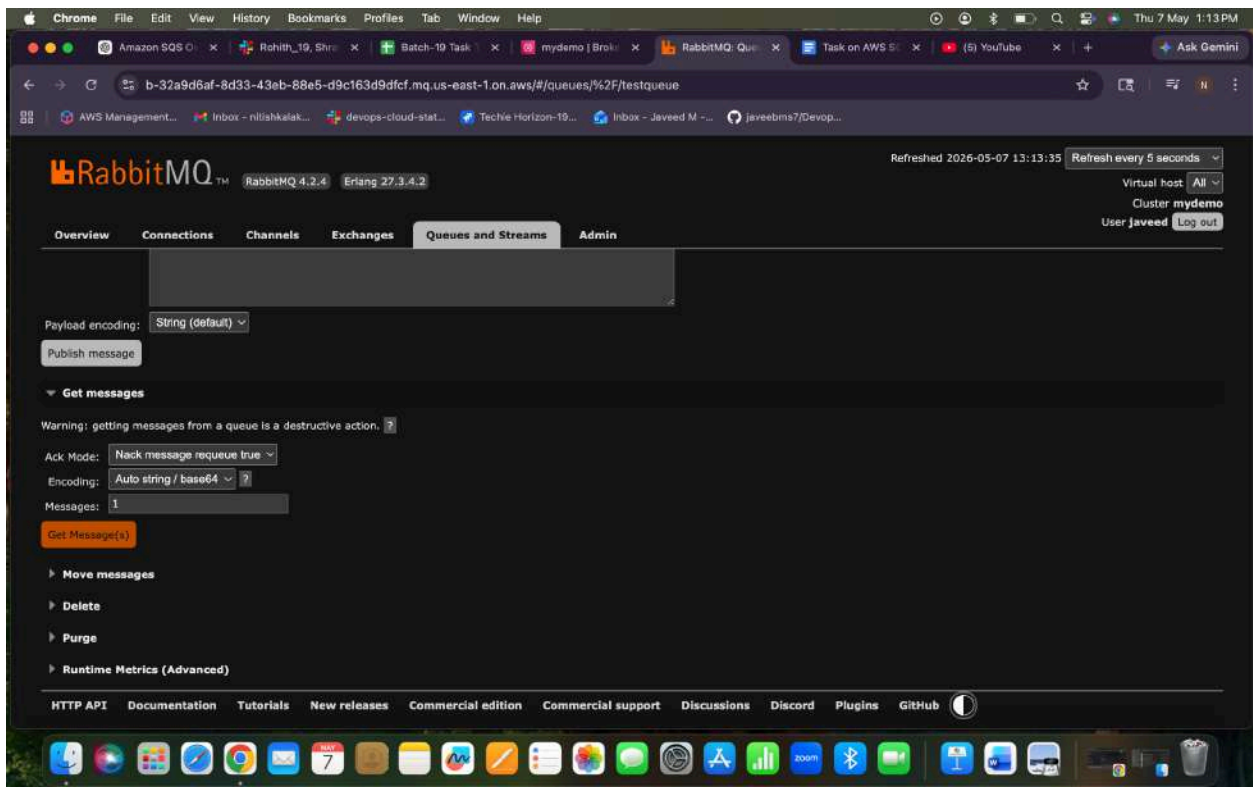
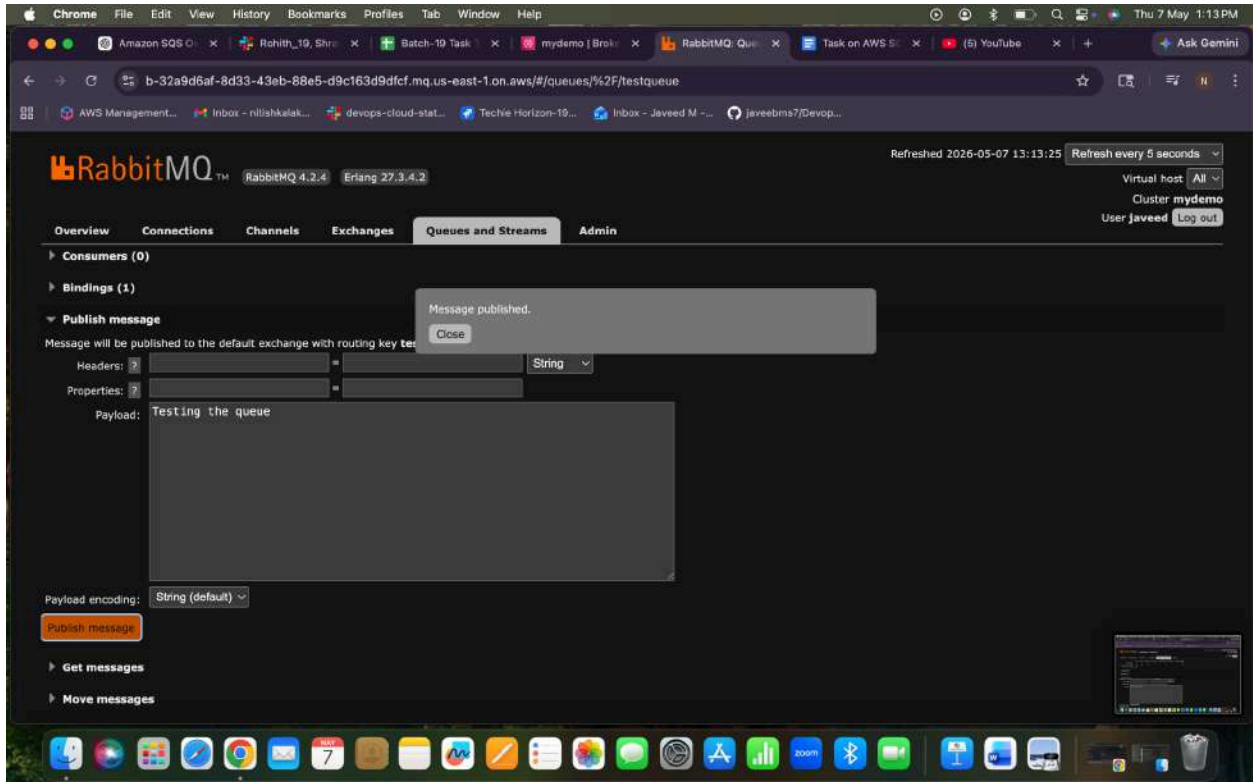
Message will be published to the default exchange with routing key **testqueue**, routing it to this queue.

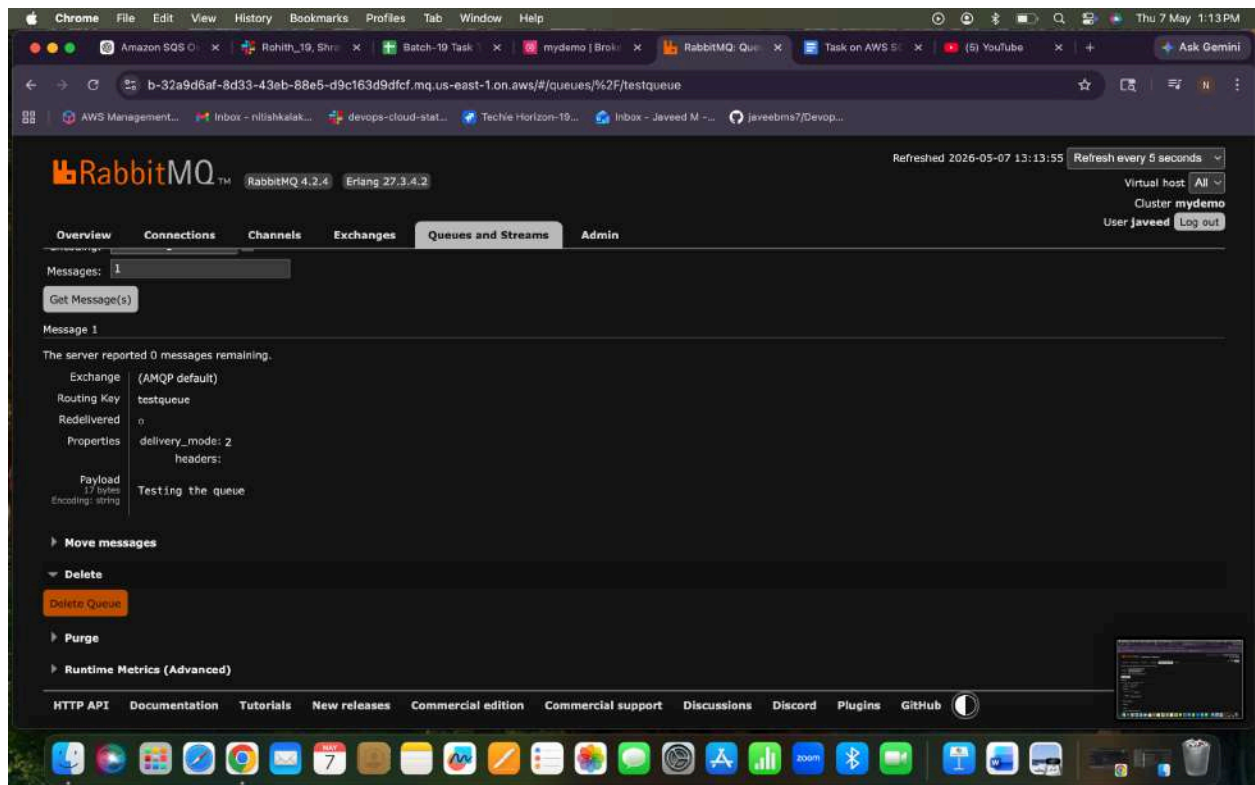
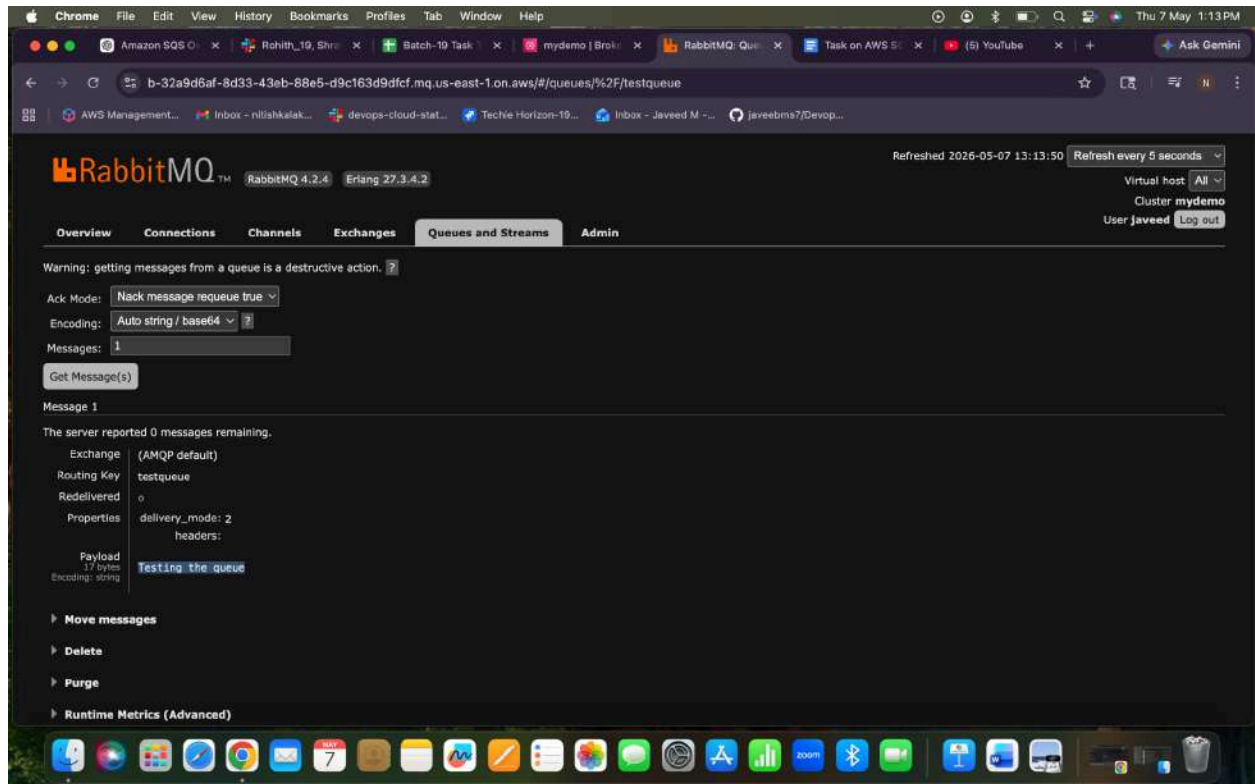
Headers: = String

Properties: =

Payload: Testing the queue

Payload encoding: String (default)





AWS Secrets Manager

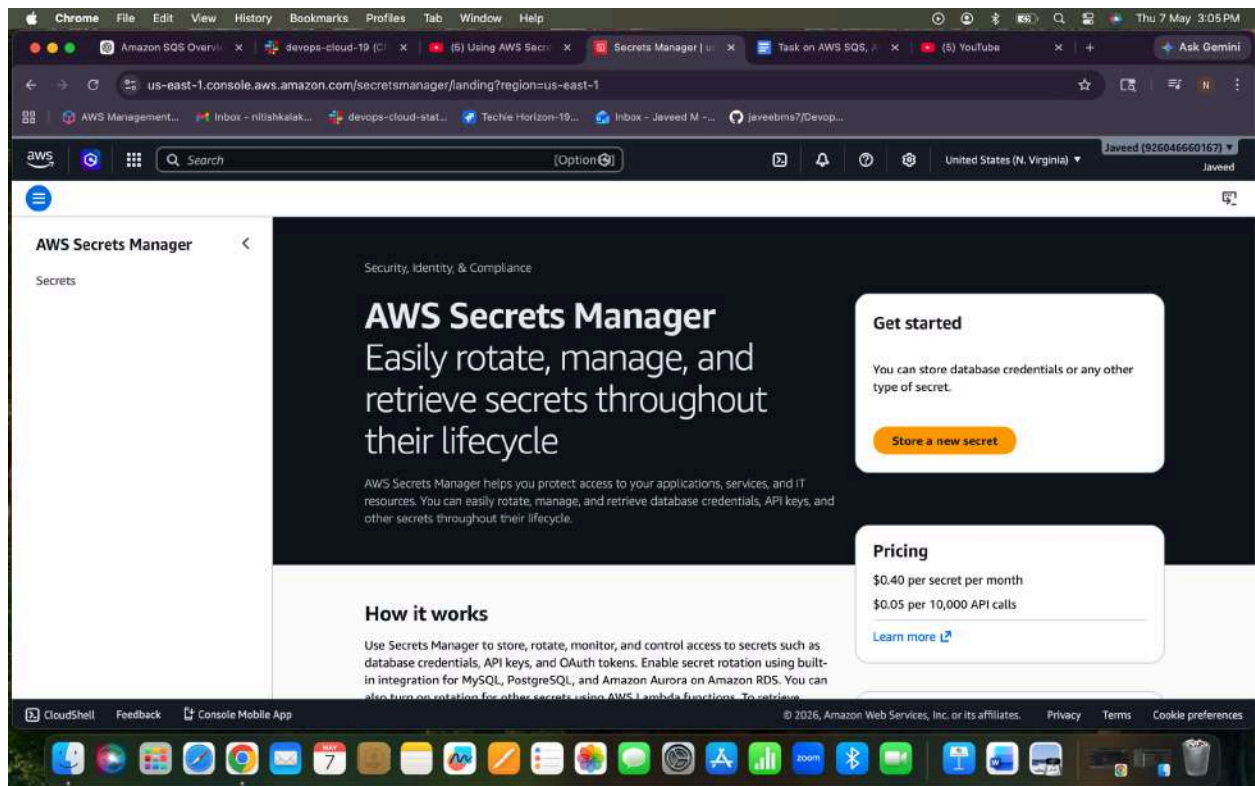
AWS Secrets Manager is a fully managed AWS service used to securely store, manage, and retrieve sensitive information such as:

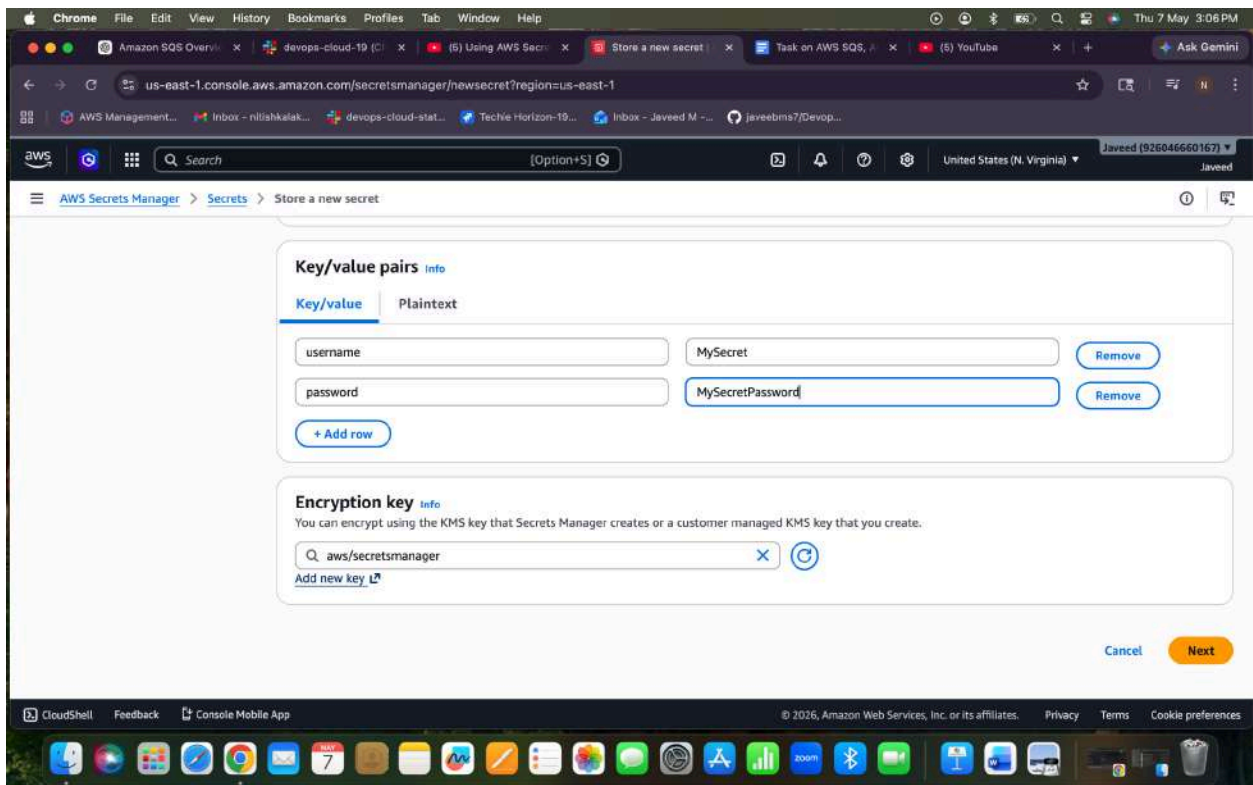
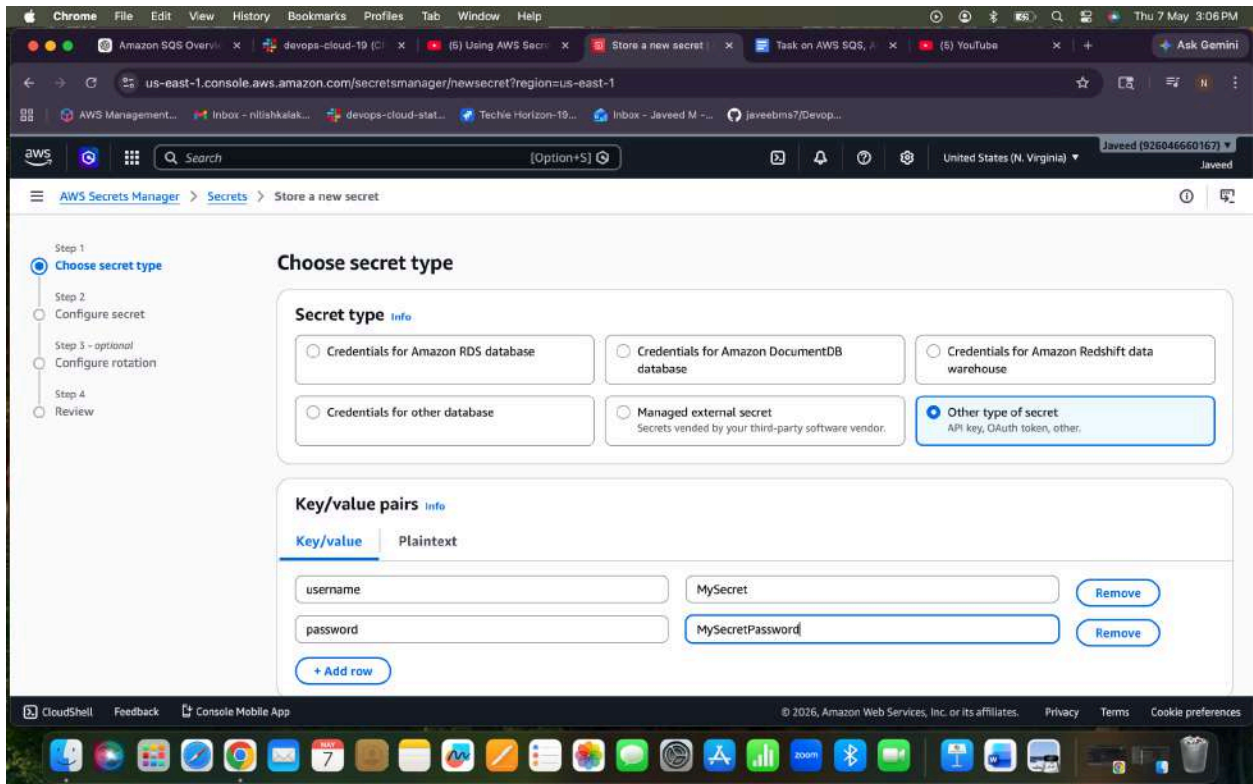
- Database passwords
- API keys
- Tokens
- SSH keys
- Application credentials

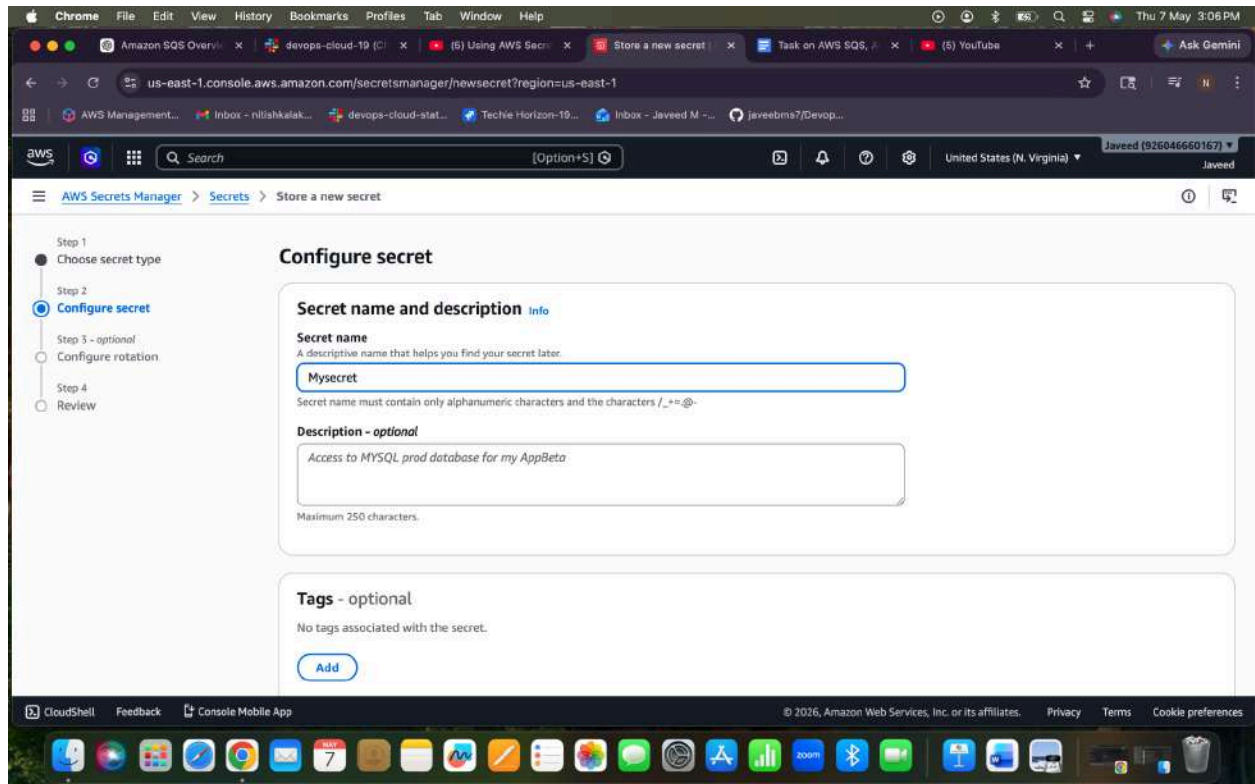
Instead of hardcoding passwords in applications or scripts, Secrets Manager stores them securely and provides controlled access

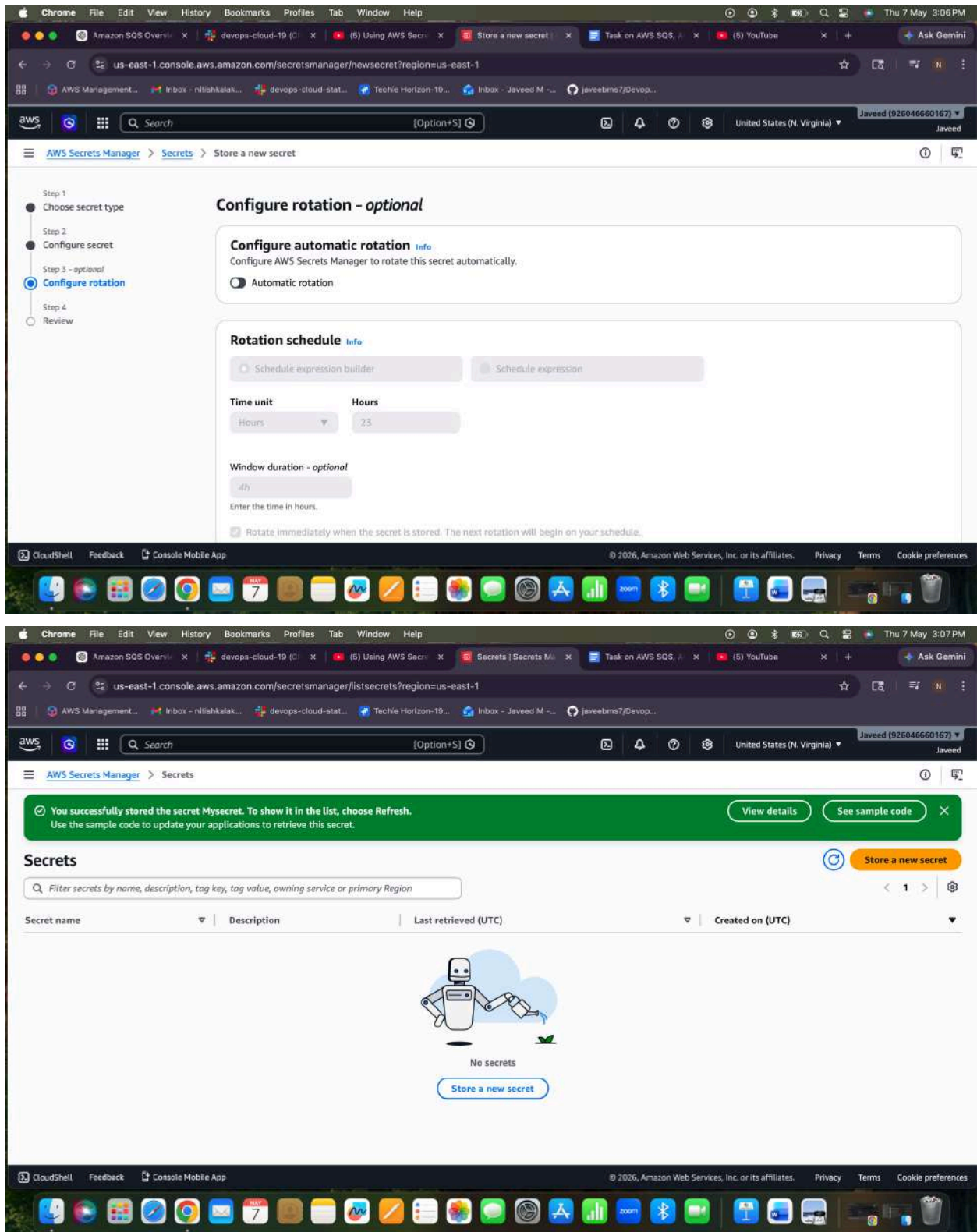
How to create

Secrets Manager → Store a new secret → choose secret type → Secret type (Different options) → Key Value pairs → Key/Value – **USERNAME/PASSWORD** → ENCRYPTION KEY → secret name & Description → Name → configure rotation (Default) → Store.



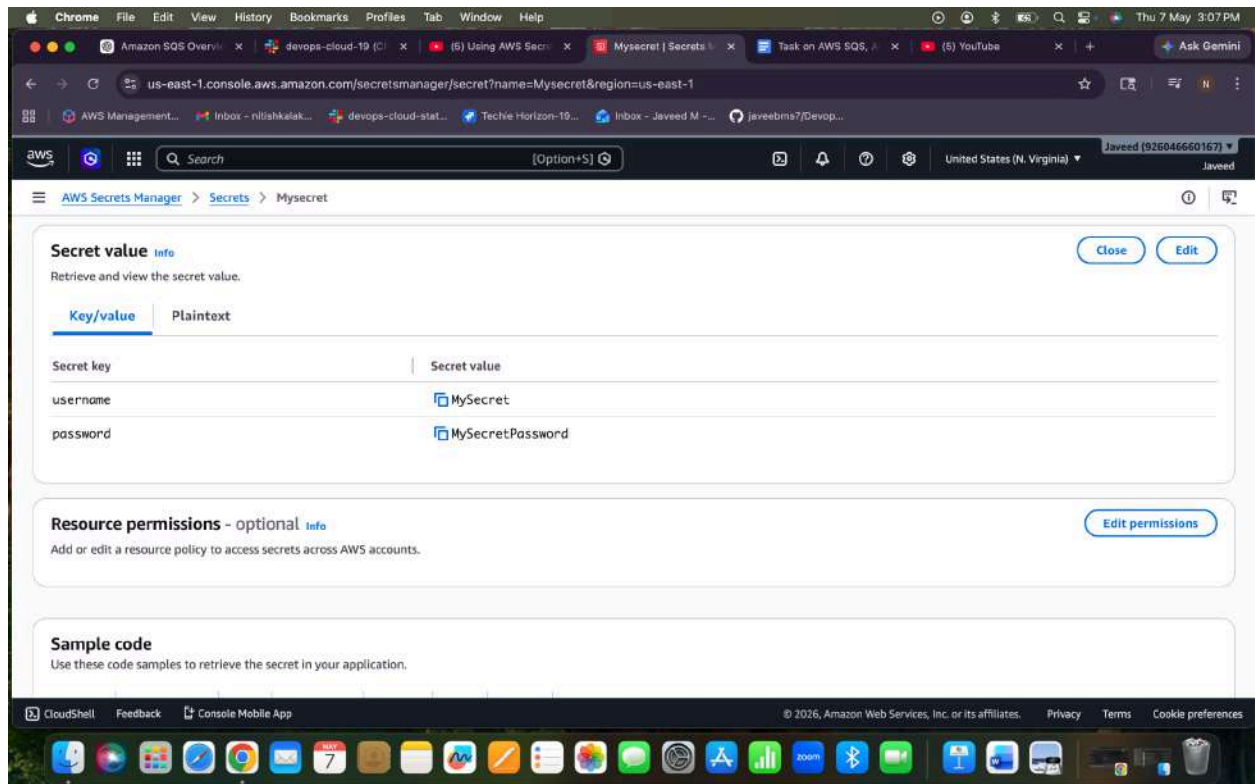
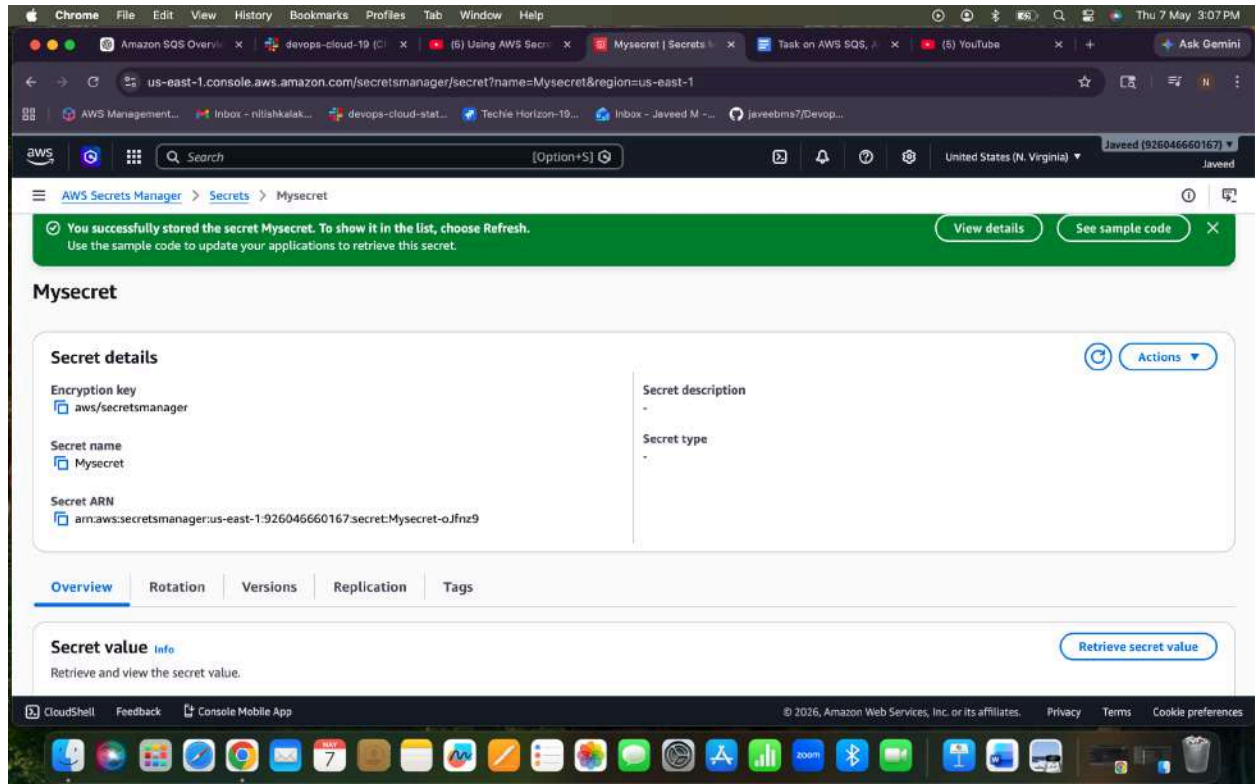






How retrieve the secret we created, from AWS GUI

Secrets Manager→Secrets→My Secret→Secret Value→ Retrieve Secret Value



If we want to check using CLI, First we need to configure AWS in our local CLI
aws configure
aws secretsmanager describe-secret --secret-id Mysecret

```
mohammedimran — -bash — 80x24
Last login: Thu May  7 10:05:54 on console

The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
[192:~ mohammedimran$ aws configure
[AWS Access Key ID [*****2IM5]:
[AWS Secret Access Key [*****rxM4]:
[Default region name [us-east-1]:
[Default output format [json]:
[192:~ mohammedimran$ aws secretsmanager describe-secret --secret-id Mysecret
Error: An error occurred (SecretsManagerException) when calling the DescribeSecret operation: Secrets Manager can't find the specified secret.
[192:~ mohammedimran$ aws secretsmanager describe-secret --secret-id Mysecret
{
  "ARN": "arn:aws:secretsmanager:us-east-1:926046660167:secret:Mysecret-oJfnz9",
  "Name": "Mysecret",
  "LastChangedDate": "2026-05-07T15:07:12.028000+05:30",
  "LastAccessedDate": "2026-05-07T05:30:00+05:30",
  "Tags": [],
  "VersionIdsToStages": {
    "71a6cc42-8fcd-486a-b157-f2442ec56de5": [
      "AWSCURRENT"
    ]
  },
  "CreateDate": "2026-05-07T15:07:11.794000+05:30"
}
[192:~ mohammedimran$
```

To get the information of secret

aws secretsmanager get-secret-value --secret-id Mysecret --version-stage AWSCURRENT

```
192:~ mohammedimran$ aws secretsmanager get-secret-value --secret-id Mysecret --version-stage AWSCURRENT
{
  "ARN": "arn:aws:secretsmanager:us-east-1:926046660167:secret:Mysecret-oJfnz9",
  "Name": "Mysecret",
  "VersionId": "71a6cc42-8fcd-486a-b157-f2442ec56de5",
  "SecretString": "{\"username\":\"MySecret\",\"password\":\"MySecretPassword\"}",
  "VersionStages": [
    "AWSCURRENT"
  ],
  "CreateDate": "2026-05-07T15:07:12.022000+05:30"
}
```

AWS Systems Manager

AWS Systems Manager is a fully managed AWS service used to manage, monitor, automate, and operate AWS resources at scale.

It helps administrators manage:

- EC2 instances
- On-premise servers
- Hybrid environments
- AWS resources

from a single centralized service.

For Using system manager we have some pre -requests like IAM Role and ec2 instance

For Creating an IAM Role:

IAM→Policies→Roles→create a Role→select trust equity→use case(ec2) →Add permissions(SSM) →Role details→Name→Create Role

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us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles

Identity and Access Management (IAM)

Roles (9) info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Role name	Trusted entities	Last activity
AWSServiceRoleForAmazonMQ	AWS Service: mq (Service-Linked Rol	-
AWSServiceRoleForAPIGateway	AWS Service: ops.apigateway (Servic	-
AWSServiceRoleForElasticLoadBalancing	AWS Service: elasticloadbalancing (S	7 days ago
AWSServiceRoleForNATGateway	AWS Service: ec2-nat-gateway (Serv	9 days ago
AWSServiceRoleForRD5	AWS Service: rds (Service-Linked Rol	1 hour ago
AWSServiceRoleForResourceExplorer	AWS Service: resource-explorer-2 (Se	17 minutes ago
AWSServiceRoleForSupport	AWS Service: support (Service-Link	-
AWSServiceRoleForTrustedAdvisor	AWS Service: trustedadvisor (Service	-
firstlambda-role-p7ok2r2v	AWS Service: lambda	14 hours ago

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us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/create

Create role

Step 1: Select trusted entity

Select trusted entity info

Trusted entity type

- ☒ AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Choose a use case for the specified service.

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Amazon SQS Overview | I syedtahquadril-10 | Instances [EC2] us-... | Create role | IAM | G... | Task on AWS SQS, ... | (5) How to create A... | + | Ask Gemini

us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/roles/create?trustedEntityType=AWS_SERVICE&selectedService=EC2&selectedUseCase=EC2

Search [Option+S]

GlobalJaveed (926046660157)Javeed

IAM > Roles > Create role

Step 1
Select trusted entity

Step 2
Add permissions

Step 3
Name, review, and create

Add permissions info

Permissions policies (1/1152) Info

Choose one or more policies to attach to your new role.

amazonssmmana

Filter by Type

All types

2 matches

< 1 >

Policy name

Type

Description

☐

AmazonSSMManagedEC2InstanceDef...

AWS managed

This policy enables AWS Systems Man...

☒

AmazonSSMManagedInstanceCore

AWS managed

The policy for Amazon EC2 Role to en...

Set permissions boundary - optional

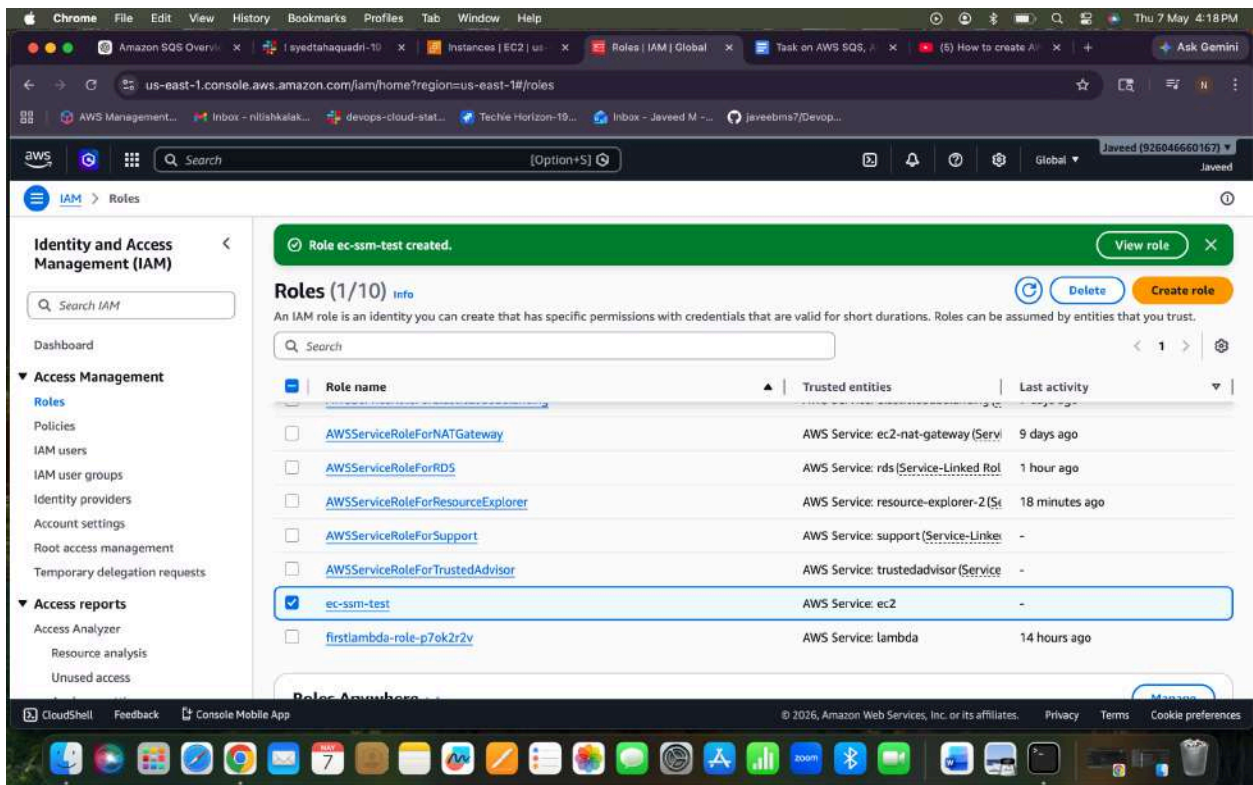
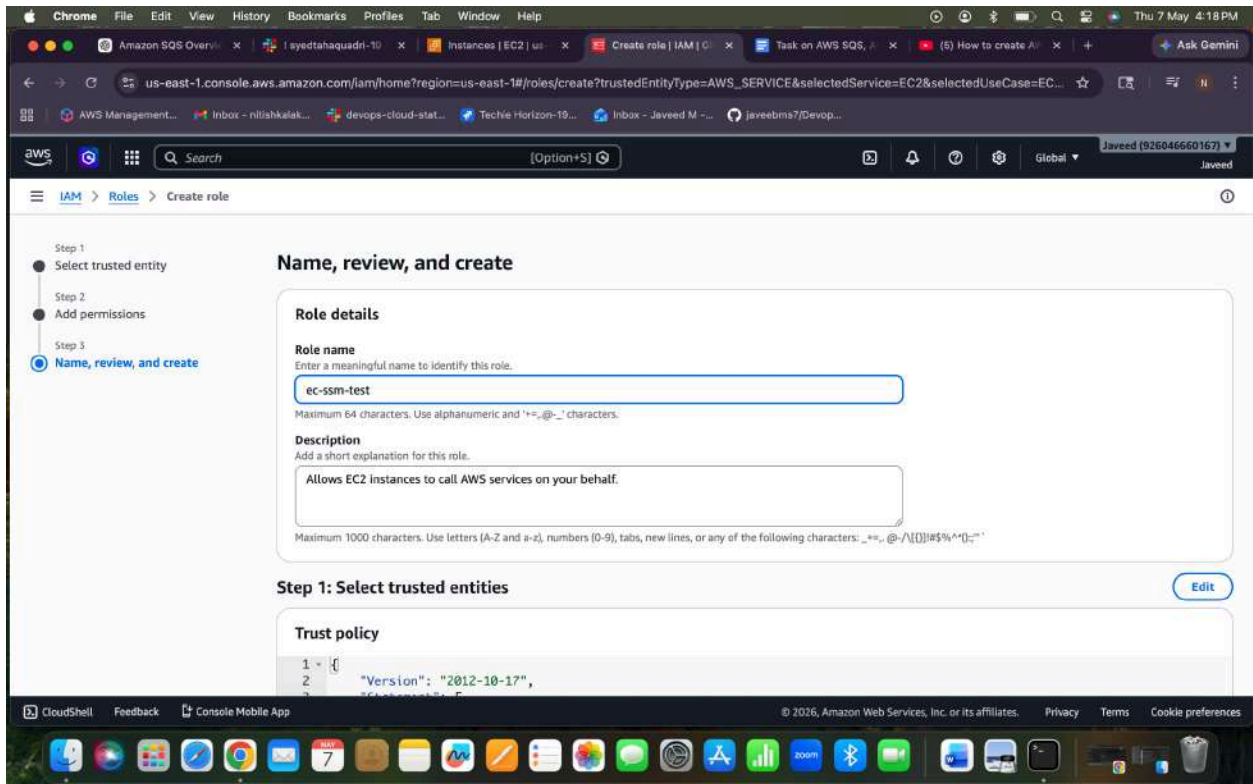
Cancel

Previous

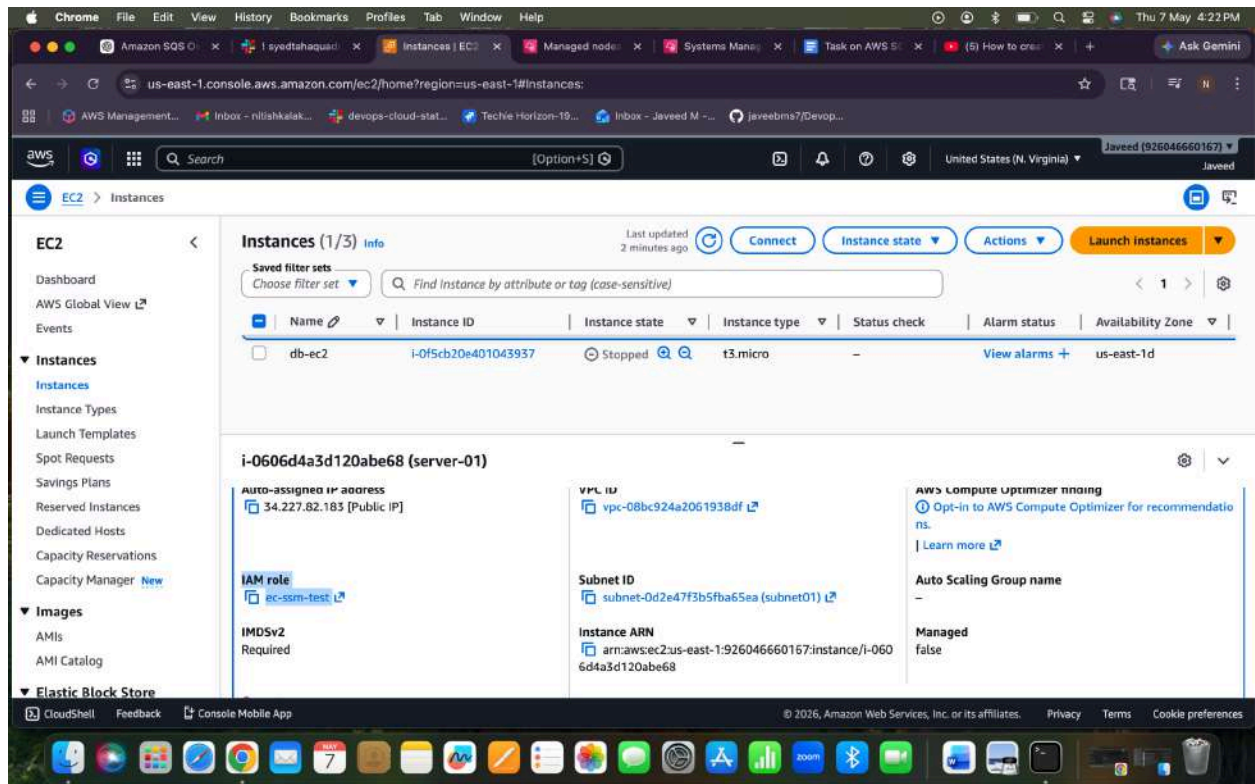
Next

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Create ec2 Instance with IAM role attached to it



Create an AWS System Manager

AWS System Manager→ Fleet Manager→Managed nodes→select
Instances→nodeactions→connect→start terminal session
Once the terminal starts we will check if we have access or not.

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Amazon SQS Overview x I saydtahaquadi-10 x Instances [EC2] us- x Home | Systems Man x Task on AWS SQS, x (5) How to create A x + Ask Gemini

us-east-1.console.aws.amazon.com/systems-manager/home?region=us-east-1#welcome

AWS Management... Inbox - nllshkalak... devops-cloud-stat... Techie Horizon-T9... Inbox - Javeed M... jareebms7/Devop...

aws Search [Option+S] United States (N. Virginia) Javeed (926046660167) Javeed

AWS Systems Manager

Manage nodes at scale in any environment

AWS Systems Manager helps you manage and operate nodes at scale on AWS, on-premises, and in hybrid and multcloud environments.

Introducing the new integrated experience

The new Systems Manager experience is an intuitive, easy to use interface to simplify node management and enhance operational efficiency. You can also set up Systems Manager across your organization. [Learn more](#)

Setting up the Systems Manager unified console performs the following actions in this account in the **us-east-1** Region. [Learn more](#)

- Enables a Default Host Management Configuration (DHMC) check to ensure that managed nodes have the permissions required for management by Systems Manager and remediates drift every 1 day.
- Collects inventory metadata from managed nodes every 12 hours.

Pricing

[Systems Manager pricing](#)

Getting started

[What is AWS Systems Manager?](#)

[Setting up AWS Systems Manager](#)

[Setting up managed nodes for AWS Systems](#)

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Amazon SQS Overview x I saydtahaquadi-10 x Instances [EC2] us- x Managed nodes | Fle x Task on AWS SQS, x (5) How to create A x + Ask Gemini

us-east-1.console.aws.amazon.com/systems-manager/fleet-manager/managed-nodes?region=us-east-1

AWS Management... Inbox - nllshkalak... devops-cloud-stat... Techie Horizon-T9... Inbox - Javeed M... jareebms7/Devop...

aws Search [Option+S] United States (N. Virginia) Javeed (926046660167) Javeed

Try out the new AWS Systems Manager unified console
The unified console makes it easier to manage nodes across your organization - whether it's EC2 instances, hybrid servers or servers running in a multi cloud environment. [Learn more](#) [Get started](#)

Systems Manager > Fleet Manager > Managed nodes

Fleet Manager

Settings Account management

You may have unmanaged Amazon EC2 instances
You can automatically configure Amazon EC2 instances as managed instances in your current account and Region by enabling Default Host Management Configuration. [Learn more](#) [Configure Default Host Management](#)

Managed Nodes (1)

Filter

Last fetched at: 4:20 PM

	Node ID	Node s...	Name	Platfor...	Operat...	Resour...	Source ID	Ping st...	Agent ...	Image ID
☐	i-0606d4a...	Running	server-01	Linux	Amazon L...	EC2 insta...	-	Online	3.3.4108.0	ami-098e...

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Chrome browser window showing the AWS Management Console. The address bar displays `us-east-1.console.aws.amazon.com/systems-manager/fleet-manager/managed-nodes?region=us-east-1`. The page title is "Fleet Manager". A notification banner states: "You may have unmanaged Amazon EC2 instances. You can automatically configure Amazon EC2 instances as managed instances in your current account and Region by enabling Default Host Management Configuration. [Learn more](#)".

The "Managed Nodes (1)" section shows a table with one node:

Node ID	Node status	Name	Platform	Operating system	Resource	Source ID	Ping status	Agent version	Image ID
i-0606d4a...	Running	server-01	Linux	Amazon Linux	EC2 Instance	-	Online	3.3.4108.0	ami-098e...

Buttons for "Start terminal session", "Node actions", "Node overview", "Connect", "Tools", and "Node settings" are visible.

Terminal window showing the command `df -h` output:

```
sh-5.2$ sudo su
[root@ip-172-31-0-179 bin]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           459M   0  459M   0% /dev/shm
tmpfs           184M  436K  183M   1% /run
/dev/nvme0n1p1  8.0G  1.7G   6.3G  21% /
tmpfs           459M   0  459M   0% /tmp
/dev/nvme0n1p128 10M   1.3M   8.7M  13% /boot/efi
tmpfs           92M   0   92M   0% /run/user/0
[root@ip-172-31-0-179 bin]#
```